

12. Answer: D

$$pV = nRT$$

$$\frac{1}{p} = \frac{1}{nRT} (V) \rightarrow \text{gradient of graph} = \frac{1}{nRT}$$

$$\text{new gradient} = \frac{1}{(2n)R(2T)} = \frac{1}{4} \left(\frac{1}{nRT} \right)$$

13. Answer: D